

Red Hat Lightspeed 1-latest

Managing system content and patch updates on RHEL systems

How to review applicable advisories and affected systems, manage system content, and remediate issues

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Abstract

This document demonstrates how to review applicable advisories and affected systems in your environment, manage system content and updates, and perform remediation using Ansible playbooks.

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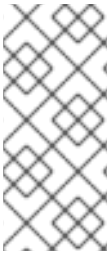
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CHAPTER 1. CONTENT PATCHING OVERVIEW

Patching leverages Red Hat software and management automation expertise to enable consistent patch workflows for Red Hat Enterprise Linux (RHEL) systems across the open hybrid cloud. It provides a single canonical view of applicable advisories across all of your deployments.

Use content patching in Red Hat Lightspeed to:

- see all of the applicable Red Hat and Extra Packages for Enterprise Linux (EPEL) advisories for your RHEL systems checking into Red Hat Lightspeed.
- patch any system with one or more advisories by using remediation plans.
- see package updates that are available for Red Hat and non-Red Hat repositories as of the last system check-in. To do so, your host must be running Red Hat Enterprise Linux RHEL 7, RHEL 8.6+ or RHEL 9 and it must have a fresh **yum** or **dnf** cache.



NOTE

- Configure role-based access control (RBAC) in Red Hat Hybrid Cloud Console > the **Settings** icon (⚙) > Identity & Access Management > User Access > Users .
- See User Access Configuration Guide for Role-based Access Control (RBAC) for more information about this feature and example use cases.

1.1. CRITERIA FOR PATCH AND VULNERABILITY ERRATA

The content patching function collects a variety of data to create meaningful and actionable errata for your systems. The insights-client collects the following data on each check point:

- List of installed packages, including name, epoch, version, release, and architecture (NEVRA)
- List of enabled modules (RHEL 8 and later)
- List of enabled repositories
- Output of **yum updateinfo -C** or **dnf updateinfo -C**
- Release version from systems with a version lock
- System architecture, for example, **x86_64**

Additionally, Red Hat Lightspeed collects metadata from the following data sources:

- Product repositories delivered by the Red Hat Content Delivery Network (CDN)
- Extra Packages for Enterprise Linux (EPEL) repositories
- Common Security Advisory Framework (CSAF)
- Vulnerability Exploitability eXchange (VEX)

Red Hat Lightspeed compares the set of system data to the collected errata and vulnerability metadata to generate a set of available updates for each system. These updates include package updates, Red Hat errata, and Common Vulnerabilities and Exposures (CVEs).



IMPORTANT

Unlike the patch service, the vulnerability service supports only official Red Hat source repositories and does not support custom repositories. Red Hat Lightspeed vulnerability can find CVEs in local mirrors of official Red Hat repositories, but only if the original Red Hat designated name is preserved. If your infrastructure uses custom or renamed Red Hat local mirror repositories, CVEs or errata from those sources will not appear in the Red Hat Lightspeed vulnerability results.

Additional resources

- Assessing and monitoring security vulnerabilities on RHEL systems
- Security > Vulnerability > CVEs

1.2. REVIEW AND FILTER APPLICABLE ADVISORIES AND SYSTEMS IN THE INVENTORY

You can see all of the applicable advisories and installed packages for systems checking into Red Hat Lightspeed.

Procedure

1. On Red Hat Hybrid Cloud Console , navigate to Content > Advisories.
2. You can also search for advisories by name using the search box, and filter advisories by:
   a. Type - Security, Bugfix, Enhancement, Unknown
   b. Publish date - Last 7 days, 30 days, 90 days, Last year, or More than 1 year ago
3. Navigate to Content > Systems to see a list of affected systems you can patch with applicable advisories. You can also search for specific systems using the search box.
4. Navigate to Content > Packages to see a list of packages with updates available in your environment. You can also search for specific packages using the search box.

1.3. SYSTEM PATCHING USING RED HAT LIGHTSPEED REMEDIATION PLANS

You can use Red Hat Lightspeed remediation plans to patch your systems.

Procedure

1. To patch your system from the Content > Advisories page in Red Hat Lightspeed:
   a. On Red Hat Hybrid Cloud Console , navigate to Content > Advisories.
   b. Click the advisory that you want to apply to affected systems. A description of the advisory, a link to view packages and errata at access.redhat.com, and a list of affected systems appear. The total number of applicable advisories from each type **Security**, **Bugfix**, and **Enhancement** against each system are also displayed. As a bulk operation, you can click the options menu located next to a system, then click **Apply all applicable advisories** to patch the system with all applicable advisories at once.

c. Alternatively, select the system(s) you want to patch with this particular advisory, then click **Plan remediation**.

d. On the Remediate with Ansible page, you can choose to modify an existing Playbook or create a new one to remediate with Ansible. Accordingly, select **Existing Playbook** and the playbook name from the drop-down list, then click **Next**. Or, select **Create new Playbook** and enter a name for your playbook, then click **Next**.

e. You will then see a summary of the action and resolution. If a reboot is required to fix the issue or risk, each system will be automatically rebooted. To disable automatic reboot, toggle the **Auto-reboot** button on the review summary panel.

f. Click **Submit**.

g. On the left navigation, click on Remediations.

h. Click on the playbook name to see the playbook details, or simply select and click **Download playbook**.

2. To patch your system from the Content > Systems page:

a. Click the **Systems** tab to see a list of affected systems. As a bulk operation, you can click the options menu located next to a system, then click **Apply all applicable advisories** to patch the system with all applicable advisories at once.

b. Alternatively, click the system you want to patch. You will see the system details and a list of applicable advisories for remediation, along with additional details such as the advisory publish date, type, and synopsis. Select the advisories you want to apply to the system, then click **Plan remediation**.

c. On the Remediate with Ansible page, you can either modify an existing Playbook or create a new one to remediate with Ansible. Accordingly, click **Existing Playbook** and select the playbook name from the drop-down list, then click **Next**. Or, click **Create new Playbook**, enter a name for your playbook, then click **Next**.

d. You will then see a summary of the action and resolution.



NOTE

If a reboot is required to fix the issue or risk, all systems in the remediation plan will be automatically rebooted. If you prefer to reboot manually after the plan has been executed, toggle the **Auto-reboot** button accordingly.

e. On the left navigation, click Automation Toolkit > Remediation Plans.

f. Click the playbook name to see the playbook details, or select and click **Download playbook**.



IMPORTANT

Review and test the recommended actions and playbooks that are available before you deploy on your Red Hat Enterprise Linux systems. Red Hat is not responsible for any adverse outcomes related to Red Hat Lightspeed recommendations or remediation plans.

1.4. UPDATE ERRATA FOR SYSTEMS MANAGED BY RED HAT SATELLITE

Red Hat Lightspeed calculates applicable updates based on the packages, repositories, and modules that a system reports when it checks in. Red Hat Lightspeed combines these results with a client-side evaluation, and stores the resulting superset of updates as applicable updates.

A system check-in to Red Hat Lightspeed includes the following content-related data:

- Installed packages
- Enabled repositories
- Enabled modules
- List of updates, which the client determines using the **dnf updateinfo -C** command. This command primarily captures package updates for non-Red Hat repositories

Red Hat Lightspeed uses this collection of data to calculate applicable updates for the system.

Sometimes Red Hat Lightspeed calculates applicable updates for systems managed by Red Hat Satellite and reports inaccurate results. This issue can occur in two ways:

- Red Hat Lightspeed shows available updates that you cannot install on the Satellite-managed system.
- Red Hat Lightspeed shows applicable updates that match what you can install on the system immediately after patching, but shows outdated or missing updates a day or two later. This can occur when the system is subscribed to RHEL repositories that have been renamed.

Red Hat Lightspeed provides an optional check-in command to generate accurate reporting for applicable updates on Satellite-managed systems. This option rebuilds the **yum** and **dnf** package caches and creates a refreshed list of applicable updates for the system.



NOTE

Satellite-managed systems are not eligible to have Red Hat Lightspeed content templates applied.

Prerequisites

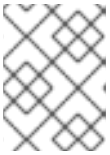
- Admin-level access to the system

Procedure

- To rebuild the package caches from the command line, enter the following command:

```
# insights-client --build-packagecache
```

The command regenerates the **dnf**/**yum** caches and collects the relevant installable errata from Satellite. The insights-client then generates a refreshed list of updates and sends it to Red Hat Lightspeed.



NOTE

The generated list of updates is equal to the output from the command **dnf updateinfo list**.

Additional resources

- Red Hat Lightspeed shows incorrect patch reporting for Satellite-managed systems
- Managing errata

1.5. AUTOMATIC CHECK-IN FOR INSIGHT-CLIENT CONFIGURATION

You can edit the **insights-client** configuration file on your system **/etc/insights-client/insights-client.conf** to rebuild the package caches automatically each time the system checks in to Red Hat Lightspeed.

Procedure

1. Open the **/etc/insights-client/insights-client.conf** file in a text editor.
2. Look in the file for the following comment:

```
#Set build_packagecache=True to refresh the yum/dnf cache during the insights-client check-in
```

3. Add the following line after the comment:

```
build_packagecache=True
```

4. Save your edits and exit the editor.

Verification

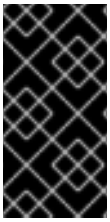
- The next time the system checks in to Satellite, insights-client executes a **yum** or **dnf** cache refresh before it collects the output of the client-side evaluation. Red Hat Lightspeed then reports the client-side evaluation output as installable updates. The evaluation output reports applicable updates based on what the Content Delivery Network (CDN) contains.

1.6. ENABLE NOTIFICATIONS AND INTEGRATIONS

You can enable the notifications service on Red Hat Hybrid Cloud Console to send notifications whenever the patch service detects an issue and generates an advisory. Using the notifications service frees you from having to continually check the Red Hat Lightspeed dashboard for advisories.

For example, you can configure the notifications service to automatically send an email message whenever the patch service generates an advisory.

Notifications for Red Hat Lightspeed services are triggered based on service-specific criteria.



IMPORTANT

For the patch service, the notification service generates notifications only about updates for the registered Red Hat Enterprise Linux systems. To receive notifications about all updates for every subscription that you have, configure the notifications service for errata events.

Enabling the notifications service requires three main steps:

- First, an Organization Administrator creates a User Access group that includes at least the **Notifications administrator** or the **RHEL administrator** role, and then adds account members to the group.
- Next, a user with the correct notifications administrator role permissions sets up behavior groups for events in the notifications service. Behavior groups specify the delivery method for each notification. For example, a behavior group can specify whether email notifications are sent to all users or just to Organization Administrators.
- Finally, users who receive email notifications from events must set their user preferences to receive individual emails for each event.

In addition to sending email messages, you can configure the notifications service to send event data in other ways:

- Using an authenticated client to query Red Hat Lightspeed APIs for event data
- Using webhooks to send events to third-party applications that accept inbound requests
- Integrating notifications with applications such as Splunk to route patch advisories to the application dashboard

In addition to sending email messages, you can configure the notifications service to send event data using an authenticated client to query Red Hat Lightspeed APIs.

Additional resources

- Configuring notifications on the Red Hat Hybrid Cloud Console
- Understanding Red Hat Lightspeed Events and Notifications in the Hybrid Cloud Console

1.7. MANAGE USER PERMISSIONS FOR RED HAT LIGHTSPEED SERVICES

Manage user permissions to control access to Red Hat Lightspeed applications. Use the User Access feature to apply role-based access control (RBAC). Red Hat provides predefined groups and roles to help Organization Administrators manage user permissions.

1.7.1. User Access overview

You can simplify access management in the Red Hat Hybrid Cloud Console by using role-based access control (RBAC). The User Access feature assigns specific permissions to roles instead of individual users, which are then inherited when roles are assigned to user groups.

You can also create custom groups and roles to provide more fine-tuned control over specific features of Red Hat Lightspeed to suit the needs of your organization.

If you are an Organization Administrator, you can use the User Access feature under **Identity & Access Management → User Access** in the Hybrid Cloud Console to:

- Control user permissions and organize roles.
- Create groups that include roles and their corresponding permissions.
- Assign users to these groups, allowing them to inherit the permissions associated with their group's roles.

Additional resources

- User Access in the Hybrid Cloud Console
- User Access Configuration Guide for Role-based Access Control (RBAC)

1.7.2. Predefined groups in User Access

The two predefined groups available in User Access, **Default access** and **Default admin access** provide different baseline privilege levels for users in your organization. You can also create custom groups to align permissions with specific personas, job functions, or teams in your organization.

The Default access group

By default, the **Default access** group is assigned many granular predefined roles, so that group members have basic visibility. Because all users in your organization are members of the **Default access** group, they inherit all permissions assigned to that group. The **Default access** group is automatically updated by Red Hat.



IMPORTANT

If your Organization Administrator modifies the **Default access** group, for example, by removing roles to restrict access to specific applications or to use the consolidated roles, the group is automatically renamed to **Custom default access**. Once converted, this group is no longer automatically updated by Red Hat.

The Default admin access group

The **Default admin access** group contains only users who have Organization Administrator permissions. This group is automatically maintained, and users and roles in this group cannot be changed.

The **Default admin access** group includes many (but not all) predefined roles that provide update and delete permissions. The roles in this group usually include **administrator** in their names.

TIP

To view the roles included in the **Default access** or **Default admin access** group, log in to the Hybrid Cloud Console, navigate to **Groups** in User Access, and select the group.

Additional resources

- User Access groups

1.7.3. Predefined roles assigned to groups

Understand how predefined roles in Red Hat Hybrid Cloud Console bundle permissions across multiple Red Hat Lightspeed applications to align with common user personas. Use predefined roles to reduce administrative effort, or create custom roles for more fine-tuned control over specific features.

The predefined roles are a starting point to help you control and manage user permissions. You can then use these roles to create custom roles that are tailored to your specific use cases and organization. For example, you can use the predefined granular roles to create custom roles that provide more fine-tuned control over specific features of Red Hat Lightspeed.

By default, Red Hat provides a set of *consolidated roles* and a set of *granular* roles in the Red Hat Hybrid Cloud Console User Access UI. The consolidated roles significantly reduce the administrative effort required to manage user permissions, while the granular roles provide more fine-tuned control over specific features of Red Hat Lightspeed.

You can use the predefined consolidated and granular roles in User Access simultaneously, but using consolidated roles can significantly reduce the administrative effort.

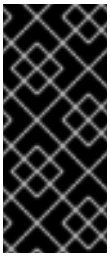
Select from the predefined consolidated roles library

The Red Hat Hybrid Cloud Console provides three predefined, consolidated User Access roles to help you manage user permissions to Red Hat Lightspeed applications and services that run on registered Red Hat Enterprise Linux systems. These roles help simplify how the Organization Administrator creates groups and permissions for various levels of access to the Red Hat Lightspeed services. If you want to reduce the administrative effort required to manage user permissions and your use case aligns with the permissions included in these roles, select from the consolidated roles library.
The consolidated roles are as follows:

RHEL viewer: The *RHEL viewer* role provides users visibility without the ability to make changes. It allows read-only access to Red Hat Lightspeed. You can view system configurations, compliance reports, inventory data, patch information, vulnerabilities, and overall resource states and activities. The only action permitted with this role is to generate activation keys.

RHEL operator: The *RHEL operator* role allows active management of your Red Hat Lightspeed environment. With this role, you can edit system configurations, inventory details, policies, and notification/integration settings. The RHEL operator role allows many of the RHEL administrator role functions, but it is restricted from editing compliance policies, content source templates, policies, or tasks. In addition, the RHEL operator role cannot execute remediation plans.

RHEL administrator: The *RHEL administrator* role provides comprehensive administrative privileges across your RHEL systems and Red Hat Lightspeed. With this role, you can manage system configurations, inventory, compliance policies, notifications, patch management, remediations, malware detection, and advisor recommendations. The role can also view and modify all vulnerability settings.



IMPORTANT

To use the consolidated roles effectively, you might need to remove the granular RHEL roles from the **Default access** group to prevent permission conflicts. This action automatically changes the name of the predefined **Default access** group to **Custom default access** group, after which, it is no longer automatically updated by Red Hat.

See *Predefined User Access roles* for a list of the roles included in the **Default admin access** group and a reference table that lists most of the predefined groups and roles that are available in the Red Hat Hybrid Cloud Console and the permissions included in each role.

Granular roles

The granular roles are specific roles for individual services that allow for fine-tuned control over specific features of Red Hat Lightspeed, for example, *Inventory Hosts administrator* or *Compliance viewer*. If you want to have more control over specific features of Red Hat Lightspeed and your use case does not align with the permissions included in the consolidated roles, use the granular predefined roles.

TIP

Across the Red Hat Lightspeed product documentation, the **Prerequisites** section for each procedure lists which predefined roles provide the permissions needed to use the features in that procedure. For example, if a procedure requires permissions to create and view remediations, the **Prerequisites** section for that procedure lists the *Remediations user* or other valid role as a recommended predefined role to use for that procedure.

Additional resources

- Preparing the Default access group for consolidated role permissions
- Predefined User Access roles

1.7.4. Check your permissions

Verify your current permissions and the roles or groups assigned to you in the Red Hat Hybrid Cloud Console. Check your permissions to troubleshoot access issues or understand your level of access to Red Hat Lightspeed applications.



NOTE

Only users with the Organization Administrator role can view the permissions of other users in the User Access settings and manage user permissions to Red Hat Lightspeed services. For more information, see the *Configure user permissions* section.

Prerequisites

- You are logged in to the Red Hat Hybrid Cloud Console.

Procedure

1. Click the **Settings** icon (⚙), then navigate to **Settings → Identity & Access Management→ My User Access**.
2. Optional: If you require additional permissions, use the Red Hat Hybrid Cloud Console Virtual Assistant to ask **"Contact my Organization Administrator"**. The assistant sends an email to the Organization Administrator on your behalf.

Verification

All of the applications that you have permissions to access are listed on this page and are grouped by product, for example, RHEL, OpenShift Container Platform, and Ansible Automation Platform.

You can also filter your permissions by application, for example, by advisor, cost management, inventory, and remediations.

Additional resources

- User Access Configuration Guide for Role-based Access Control (RBAC)
- Procedures for configuring User Access

1.7.5. Configure user permissions

If you are an Organization Administrator, you can view and manage user permissions for all users in your organization. Control access to Red Hat Lightspeed and other Red Hat Hybrid Cloud Console services through the User Access interface.



IMPORTANT

If you are not an Organization Administrator, you are unable to complete this task. However, you can check your own permissions for different applications by navigating to **Settings → Identity & Access Management→ My User Access**. Contact your Organization Administrator to request more permissions.

Prerequisites

- You have logged in to the Red Hat Hybrid Cloud Console as an Organization Administrator, or you have the required administrator User Access role permissions.

Procedure

1. Click the **Settings** icon (⚙), then navigate to **Settings → Identity & Access Management→ User Access**.

Verification

From here, you can create and manage:

- **Settings → Identity & Access Management→ User Access → Roles** to determine permissions to Red Hat Lightspeed services and features
- **Settings → Identity & Access Management→ User Access → Groups** to include one or more roles to align with a specific persona, job function, or team in your organization
- **Settings → Identity & Access Management→ User Access → Users** and their assignment to groups to inherit permissions from the roles assigned to those groups

Additional resources

- User Access Configuration Guide for Role-based Access Control (RBAC)
- Procedures for configuring User Access
- Simplifying access management on Hybrid Cloud Console for Red Hat Lightspeed with predefined roles (Red Hat Knowledgebase)

1.7.6. User Access roles for permissions to content templates and patch updates

Understand the predefined roles that control access to content templates and patch features in Red Hat Lightspeed. Use these role definitions to assign appropriate permissions to users based on their responsibilities.

The following roles enable standard or enhanced access to the content template and patch features:

Table 1.1. Permissions provided by the User Access roles

User Access role	Grants permissions to ...	Included in the Default access group
Content Template administrator	• Do any available operations on any content template or patch resource.	
Content Template viewer	• Read any content template or patch resource.	X
RHEL administrator	• Do everything that a **RHEL operator** can do. • Administer RHEL system configs, inventory, compliance, notifications, patch management, execute remediation plans, malware detection, and advisor. • View and modify vulnerability settings.	
RHEL operator	• Do everything that a **RHEL viewer** can do. • Edit system configs, inventory, policies, notifications, and integrations. • View compliance reports, patch info, malware detections, and recommendations. • Create remediation plans, manage stale data, and change vulnerability settings.  **NOTE** The RHEL operator role is restricted from editing compliance policies, content source templates, policies, or tasks. Also, the RHEL operator role cannot execute remediation plans.	

User Access role	Grants permissions to ...	Included in the Default access group
RHEL viewer	● Read all available data across Red Hat Lightspeed services and features. ○ View system configs, compliance reports, inventory data, patch info, vulnerabilities, and more to observe the state of resources or activities.  **NOTE** Cannot perform actions other than generating activation keys.	

CHAPTER 2. USE CONTENT TEMPLATES TO APPLY SYSTEM PATCHES

You can filter the errata with content templates and apply them to registered systems in the Hybrid Cloud Console. The templates provide filtering and control on the systems. You also can use the same set of content templates to build custom images in Red Hat Lightspeed.

Content templates include product and architecture attributes. These attributes, and a date that you select, create a defined set of packages and errata. You can use these sets to build images, report on patching status, and control the content delivered to your registered systems. All content templates include base RHEL repositories, and you can choose to include additional Red Hat and custom repositories.

Repository snapshots capture the state of your repository at a point in time. Repositories are collections of Red Hat Package Manager files (RPMs) and metadata.

A content template consists of a set of repository snapshots that have the same version of Red Hat Enterprise Linux and the same system architecture. You select the snapshots to ensure that your systems have access only to specific package versions. Using the content templates, you can establish date-based patch baselines for your systems, and then use **yum** or **dnf** to perform updates to the systems.

2.1. ABOUT SNAPSHOTS

Snapshots are daily reference copies of the state of your repository at specified points in time. Red Hat Lightspeed takes snapshots of the common RHEL repositories and your custom repositories.

You can view the list of snapshots for the repositories that you select, and the changes that occur in time between one snapshot and the next. For example, you can see when new packages have been added between snapshots.

2.2. ABOUT CONTENT TEMPLATES

A content template consists of an explicit set of packages and errata that can be installed or updated on any associated system. It contains a set of repository snapshots that have the same RHEL product/version, system architecture, and template creation date.

When you associate a system with a content template, Red Hat Lightspeed overwrites the repository definitions on the system to point to the repository snapshots in the template. These are the same repository snapshots that appear on the Repositories page in the Hybrid Cloud Console. Once you associate a system with a content template, any **yum** or **dnf** commands that you run on that system are constrained to the updates included in those repository snapshots.

For example, you can create or edit a content template to contain only snapshots of Red Hat Enterprise Linux 9 for **x86_64** repositories. You can use that content template to apply consistent patches to other systems in your inventory that match the product/version and architecture. Once you associate a content template with a set of systems, you can run **dnf** at the command line to easily perform consistent updates to all systems associated with that template.

Systems in your inventory that are not associated with that template do not receive the update packages and errata.

Systems that you can associate with a content template must meet the following criteria:

- Must run RHEL 8 or later. The RHEL version must match the version specified in the template.
- Registered with Red Hat Lightspeed, but not managed by Red Hat Satellite or Red Hat Update Infrastructure (RHUI).
- Must have a system architecture that matches that specified in the template.
- Must not have a release version set. To ensure that the release version is not set, run the following command as root:

```
subscription-manager release --unset
```

Additional resources

- Managing repositories to build your customized operating systems

2.3. PATCH RHEL SYSTEMS WITH CONTENT TEMPLATES

Content templates help you to apply consistent updates to all the Red Hat Enterprise Linux systems assigned to the same template. This approach separates content management from remediation and allows for more flexibility in patching. You can even use automation tools, such as Red Hat Ansible Automation Platform, to perform patching operations for you.

To use content templates, you must be logged in to the Red Hat Hybrid Cloud Console as the Organization Administrator or a member of a User Access group with at least one of the following role assignments:

- **RHEL administrator**
- **Repositories administrator** (read, write, upload) and **Content Template administrator** (read repositories, read and write templates)

You can view your permissions in **your profile > My User Access > Users** Click your account name. The list of groups for which you have permission displays.

You can see the permissions you have within a group when you click the group name. The group page displays and shows the types of permissions available (for example, read and write).

Additional resources

- Predefined User Access roles
- User Access configuration guide for role-based access control (RBAC)

2.3.1. How patching works with content templates

The workflow for patching with content templates has the following general overview steps:

1. Create the content template in Red Hat Lightspeed. The template includes the RHEL version, system architecture, template creation date, and other criteria. Systems that use the content template must match these criteria.
2. Assign systems to the template. The template points the repositories to snapshots for system updates. Note that the systems can only receive updates through the content template after you assign them to the template.

3. When a system checks in to Red Hat Lightspeed, it can discover installable updates. The updates match the criteria in the content template. To ensure consistency, all the systems associated with a template discover the same updates.



IMPORTANT

A system can be assigned to only one content template at a time.

2.4. MANAGE CONTENT TEMPLATES

A content template is a set of repository snapshots you have selected to ensure your systems only have access to specific package versions. Using the content templates, you can establish date-based patch baselines for your systems and perform updates of your systems using **yum** or **dnf**.

2.4.1. Create a content template

You can create a content template to ensure your system retains access to specific package versions.



IMPORTANT

Content templates currently support only major RHEL versions (RHEL 8, 9, or 10). When you create or edit a content template, select a major RHEL version from the drop-down menu for OS Version. Extended support (EUS) subscriptions can show minor RHEL versions in the drop-down menu. However, content templates do not support minor RHEL versions at this time.

Prerequisites

- You are logged in to the Red Hat Hybrid Cloud Console as the Organization Administrator or a user who is a member of a User Access group with at least one of the following role assignments:
  - **RHEL administrator**
  - **Repositories administrator** (read, write, upload) and **Content Template administrator** (read repositories, read and write templates)
- The custom repositories that you want to include in the template have snapshots.

Procedure

1. On the Red Hat Hybrid Cloud Console, navigate to **Content → Templates**.
2. Click **Create template**. The **Create content template** wizard opens.
3. In the **Content** section complete the following:
   a. On the **OS and architecture** page, select the **Release stream**, **Operating system version**, and **Architecture** of your system, and click **Next**.

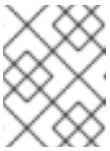


IMPORTANT

The **Release stream** is available only if you have an extended support subscription. For more information about the release streams, see Red Hat Enterprise Linux Life Cycle.

b. On the **Red Hat repositories** page, select repositories you want to include in the template. Click **Next**.

c. On the **Custom Repositories** page, select repositories you want to include in the template. Click **Next**.



NOTE

Only repositories with enabled snapshot functionality are displayed in the list of repositories.

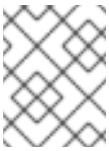
4. On the **Set snapshot date** page, select:

Use latest content

To use the latest content from the repository during image building. Click **Next**.

Use a snapshot

To use a repository snapshot. Select the date and then click **Next**.



NOTE

If your repository does not have the required snapshot, it chooses the earliest possible snapshot to the date of your choice.

5. On the **Detail** page, enter a name for your template. Optionally, add a description of your template. Click **Next**.

6. On the **Review** page, verify the details about the template and click **Create template and add to systems**. The **Assign template to systems** wizard opens.
Optional: Click **Create template only** if you want to assign this template to systems later.

7. In the **Assignment method**, select **Via system list**, and select all the systems you want to assign the template to. Click **Save**.
This change updates the **/etc/yum.repos.d/redhat.repo** file on the selected systems. All changes affect your systems every four hours. If you want this change to affect your systems immediately, you must manually refresh **subscription-manager** on the selected system:

```
# subscription-manager refresh
```

If you choose a different assigning method, follow the commands that each method provides on the systems you want to assign the template to.

Verification

1. Go to **Services → Red Hat Enterprise Linux→ Content → Templates** and verify your template appears in the list of templates.

2. On your system, check that the **/etc/yum.repos.d/redhat.repo** file contains **https://cert.console.redhat.com**.

Next steps

- You can edit your content template, assign your template to systems, and delete it. For more details, see Editing a content template .

Additional resources

- How to use content templates in Red Hat Lightspeed
- Managing repositories to build your customized operating systems

2.4.2. Edit a content template

You can change the snapshot date for your content template, assign your content template to more systems, and delete your content template.



IMPORTANT

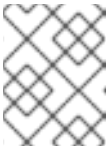
Content templates currently support only major RHEL versions, such as RHEL 8, 9, or 10. When you create or edit a content template, select a major RHEL version from the drop-down menu for OS Version. Extended support (EUS) subscriptions can show minor RHEL versions in the drop-down menu. However, content templates do not support minor RHEL versions at this time.

Prerequisites

- You are logged in to the Red Hat Hybrid Cloud Console as the Organization Administrator or a user who is a member of a User Access group with at least one of the following role assignments:
  - **RHEL administrator**
  - **Repositories administrator** (read, write, upload) and **Content Template administrator** (read repositories, read and write templates)
- The custom repositories that you want to include in the template have snapshots.

Procedure

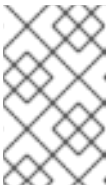
1. On the Red Hat Hybrid Cloud Console, navigate to **Content → Templates**.
2. Select the content template you want to edit and choose an action from the following options:
   - To edit repositories, snapshot date, and the name of your content template, go to **Actions → Edit** and follow the **Edit content template** wizard.



NOTE

If you edit a snapshot date for the template, it sets a new baseline for patching, and the systems start using the new snapshot.

- To assign this template to more systems, go to **Services → Red Hat Enterprise Linux→ Content → Templates**, open the template you want to assign to more systems, click **Assign template to systems**, and select the systems you want to assign this template to. Click **Save**.



NOTE

If you do not have systems to which you can apply your template, you can click **Register and assign via API**, and follow the commands the console provides.

- To delete the content template, go to **Actions → Delete**.

Additional resources

- How to use content templates in Red Hat Lightspeed
- Managing repositories to build your customized operating systems

2.5. VIEW THE AVAILABLE ADVISORIES FOR A CONTENT TEMPLATE

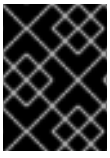
You can view and filter the available advisories for a content template. You can decide on the best plan for patching your systems based on this information.

Procedure

1. Navigate to **Content > Templates** and select a content template from the list. The information page for the template displays.
2. Select **Advisories** under the **Content** tab. The list contains all advisories that apply to systems associated with that content template.
3. **Optional:** To filter advisories, select the criterion you want to use from the drop-down menu (**Name/Synopsis**, **Type**, or **Severity**), and then use the second drop-down to select the filter. For example, you can view advisories by severity, and then filter the list to show only the Critical and Important advisories.

2.6. INSTALL UPDATES ON RHEL SYSTEMS FROM A CONTENT TEMPLATE

Content templates help you to apply consistent updates to all the systems assigned to the same template. This approach separates content management from remediation, and allows for more flexibility in patching. You can use automation tools to perform patching operations on your systems.



IMPORTANT

RHEL 8 supports both the **dnf** and **yum** update commands. RHEL 9 and later support only **dnf** for updates.

Prerequisites

- You have root access to the system that you want to update.

- You have set up your repositories and snapshots.
- You have created content templates and assigned systems to them.
- Updates are available for systems associated with the content template.

Procedure

1. At the command line, type **dnf update**.

   ```
   # dnf update
   ```

   This installs the updates to the system. The system receives only the updates associated with the content template. For example, if your content template has systems running RHEL 9, then the updates install only on RHEL 9 systems. Systems in your inventory that have different operating system versions or system architectures remain unaffected.

2. Repeat the update on each of the other systems associated with the same content template.